



SDN and NFV Overview



Foreword

- The open ecosystem of the computing industry brings booming development of multiple fields, such as Commercial Off-the-Shelf (COTS), operating system, virtualization, middleware, cloud computing, and software applications. The network industry is also seeking transformation and development. Software Defined Networking (SDN) and Network Functions Virtualization (NFV) are mainly used.
- This course aims to help engineers understand the development of SDN and NFV and introduce Huawei SDN and NFV solutions.



Objectives

- Upon completion of this course, you will be able to:
 - Describe the development of SDN and NFV.
 - Understand basic principles of OpenFlow.
 - Understand Huawei SDN solution.
 - Understand the standard NFV architecture.
 - Understand Huawei NFV solution.

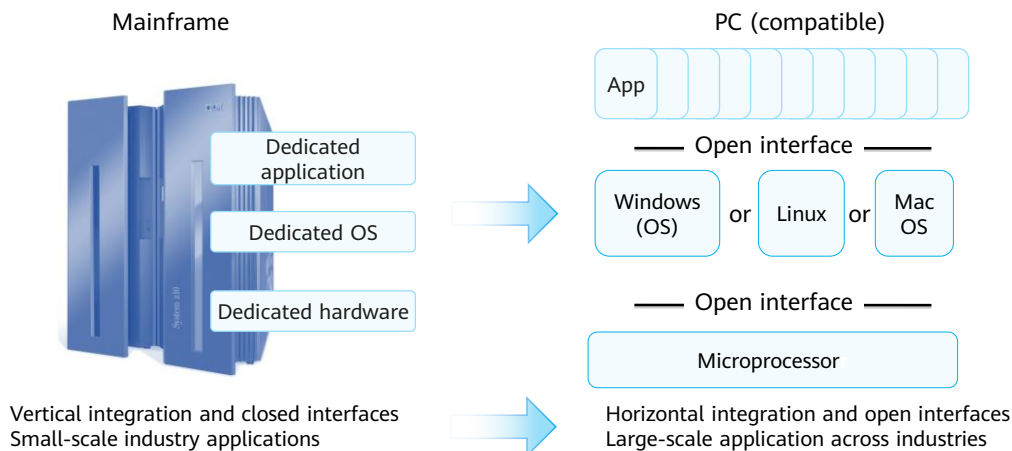


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Evolution of the Computer Era



- In 1964, IBM spent US\$5 billion on developing IBM System/360 (S/360), which started the history of mainframes. Mainframes typically use the centralized architecture. The architecture features excellent I/O processing capability and is the most suitable for processing large-scale transaction data. Compared with PCs, mainframes have dedicated hardware, operating systems, and applications.
- PCs have undergone multiple innovations from hardware, operating systems, to applications. Every innovation has brought about great changes and development. The following three factors support rapid innovation of the entire PC ecosystem:
 - Hardware substrate: The PC industry has adapted a simple and universal hardware base, x86 instruction set.
 - Software-defined: The upper-layer applications and lower-layer basic software (OS and virtualization) are greatly innovated.
 - Open-source: The flourishing development of Linux has verified the correctness of open source and bazaar model. Thousands of developers can quickly formulate standards to accelerate innovation.



Network Industry Development: Implications from the IT Industry

- The transformation of the IT industry has triggered the thinking of the network industry. The industry has proposed the SDN concept and has made attempts to put SDN into commercial use, aiming to make networks more open, flexible, and simple.

Computing Industry Openness Promotes Ecosystem Development

Cloud service	 ECS	 EVS	Comprehensive cloud services
Database			
Middleware			
OS			
Virtualization			
Server, storage device, PC	 GaussDB	Various virtualization technologies, operating systems, middleware, database software, etc.	
Universal hardware	 openEuler	FusionSphere	
	 x86/ARM server	Storage array	PC ...
	 x86/ARM chip	Memory disk	Hard ...

What About Network Industry Changes



Network application

SDN controller

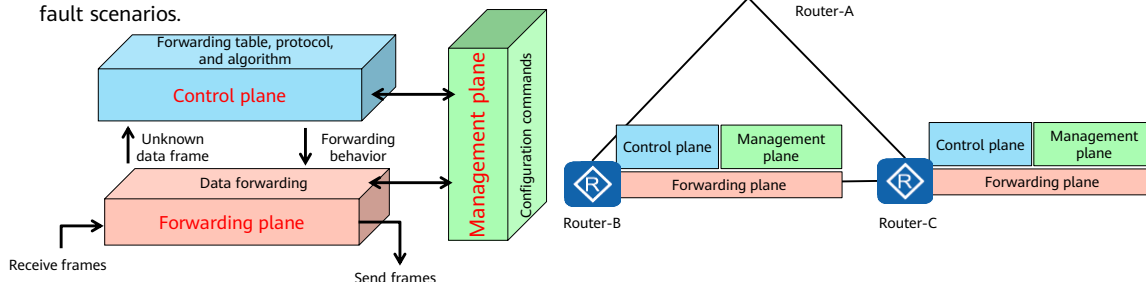
Hardware network device

- Does the network industry build a hierarchical and open ecosystem according to the computing industry?



Current Situation of the Network Industry: Typical IP Network - Distributed Network

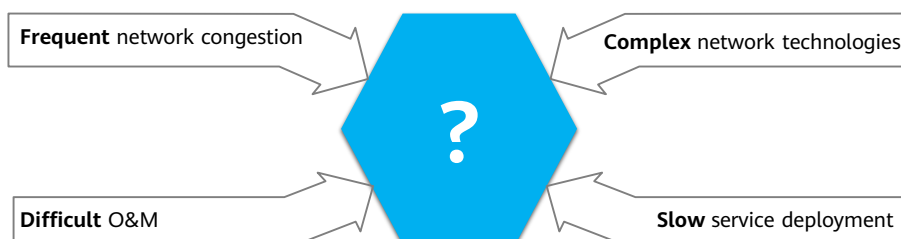
- The typical IP network is a distributed network with peer-to-peer control. Each network device has independent forwarding, control, and management planes. The control plane of a network device exchanges packets of a routing protocol to generate an independent data plane to guide packet forwarding.
- The advantage of a typical IP network is that network devices are decoupled from protocols, devices from different vendors are compatible with each other, and network convergence is ensured in fault scenarios.



- The switch is used as an example to describe the forwarding plane, control plane, and management plane.
- Forwarding plane: provides high-speed, non-blocking data channels for service switching between service modules. The basic task of a switch is to process and forward various types of data on its interfaces. Specific data processing and forwarding, such as Layer 2, Layer 3, ACL, QoS, multicast, and security protection, occur on the forwarding plane.
- Control plane: provides functions such as protocol processing, service processing, route calculation, forwarding control, service scheduling, traffic statistics collection, and system security. The control plane of a switch is used to control and manage the running of all network protocols. The control plane provides various network information and forwarding query entries required for data processing and forwarding on the data plane.
- Management plane: provides functions such as system monitoring, environment monitoring, log and alarm processing, system software loading, and system upgrade. The management plane of a switch provides network management personnel with Telnet, web, SSH, SNMP, and RMON to manage devices, and supports, parses, and executes the commands for setting network protocols. On the management plane, parameters related to various protocols on the control plane must be pre-configured, and the running of the control plane can be intervened if necessary.
- Some Huawei series products are divided into the data plane, management plane, and monitoring plane.



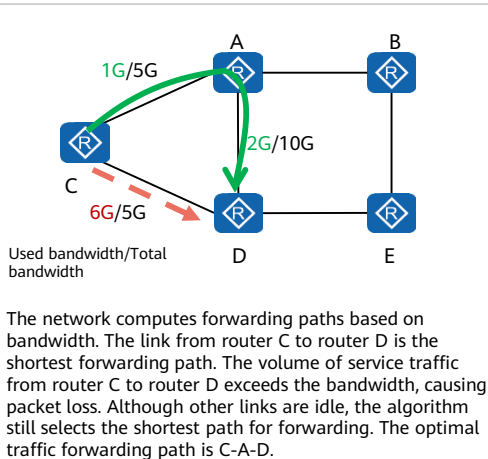
Thinking in the Network Field: Problems Faced by Typical Networks





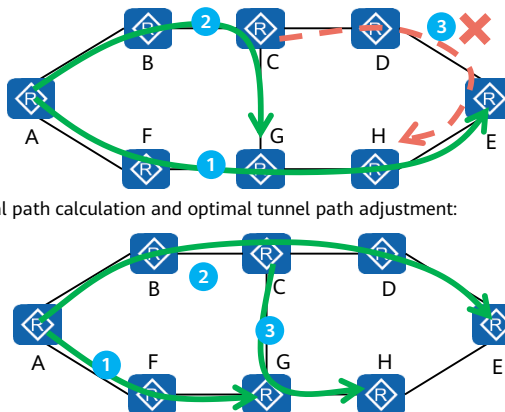
Frequent Network Congestion

Problem and Solution of Bandwidth-based Route Selection



Problem and Solution of Tunnel Establishment Based on Fixed Sequence

Tunnels are established in sequence: 1. A-E; 2. A-G; 3. C-H. Tunnel 3 fails to be established due to insufficient bandwidth.

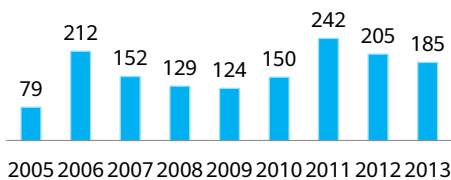




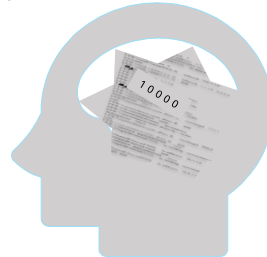
Complex Network Technologies

Many network protocols: Network technology experts need to learn many RFCs related to network devices. Understanding the RFCs takes a long time, and the number of RFCs is still increasing.

RFC increase trends



Difficult network configuration: To be familiar with devices of a specific vendor, you need to master tens of thousands of commands. Additionally, the number of commands is still increasing.





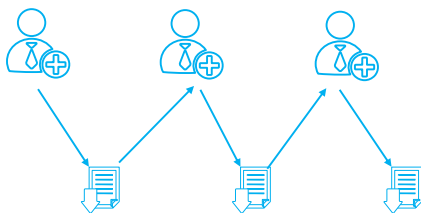
Difficulty in Locating and Analyzing Network Faults

Difficult to Spot Faults

Manual fault identification

Manual packet obtaining for locating faults

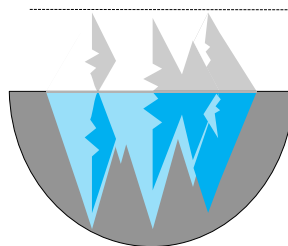
Manual fault diagnosis



- Traditional O&M networks rely on manual fault identification, location, and diagnosis.
- More than 85% of network faults are found only after service complaints. Problems cannot be proactively identified or analyzed.

Difficult to Locate Faults

Abnormal flows account for 3.65% of all flows on the network.



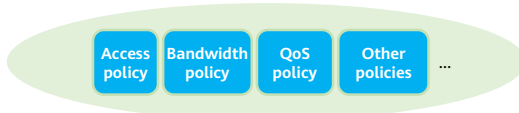
The network faults that are found upon user complaints are just the tip of the iceberg.

- Traditional O&M only monitors device indicators. Some indicators are normal, but user experience is poor. There is no correlated analysis of users and networks.
- According to data center network (DCN) statistics, it takes 76 minutes to locate a fault on average.



Slow Network Service Deployment

Network policy



Complex and inflexible network policy changes:
Network policies cannot be defined by user. Policy changes are complex and cannot be flexibly adjusted.

IP address-based, fixed location, and CLI-based configuration

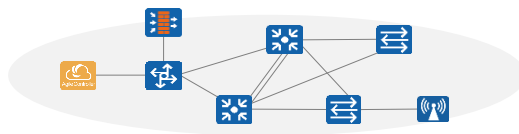
Service network



Long service deployment period:
New service deployment involves E2E device configuration modification.

End-to-end configuration using commands

Physical network



Low physical network deployment efficiency:
The physical network does not support zero touch provisioning (ZTP).

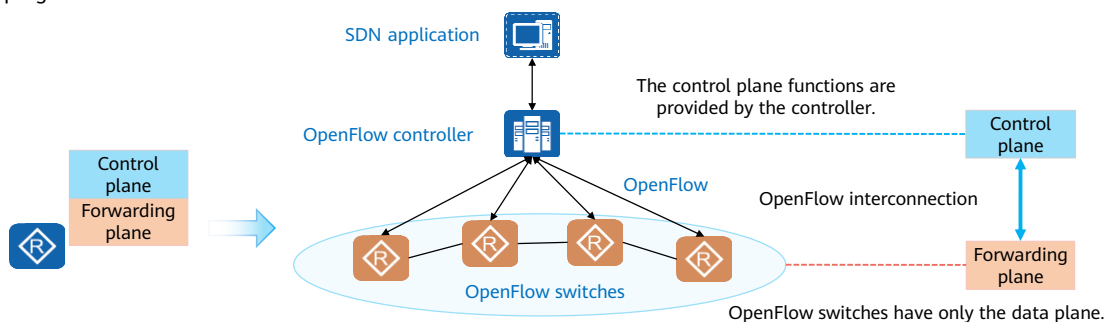
Command line-based configuration by device

- Vision of network service deployment:
 - Free mobility based on network policies, regardless of physical locations
 - Quick deployment of new service
 - ZTP deployment on the physical network
 - Plug-and-play of devices



SDN Origin

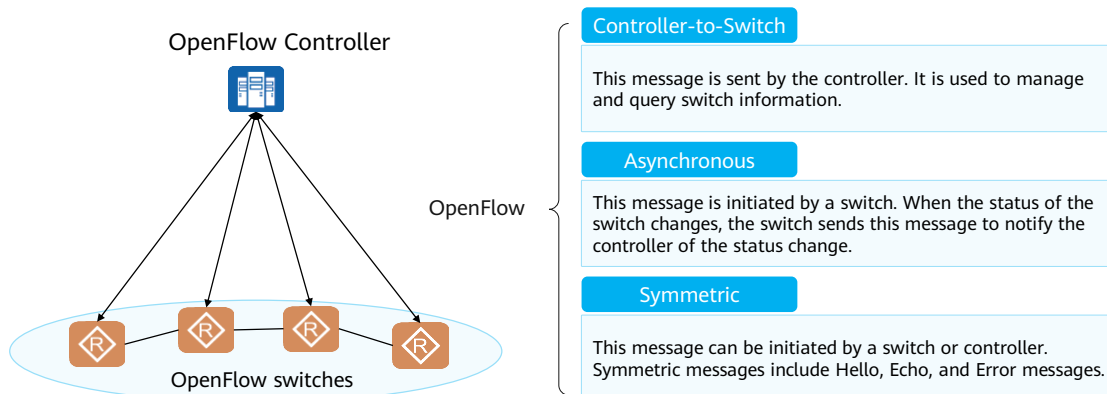
- SDN was developed by the Clean Slate Program at Stanford University as an innovative new network architecture. The core of SDN is to separate the control plane from the data plane of network devices to implement centralized control of the network control plane and provide good support for network application innovation.
- SDN has three characteristics in initial phase: forwarding-control separation, centralized control, and open programmable interfaces.





Basic Concepts of OpenFlow

- OpenFlow is an SBI protocol between a controller and a switch. It defines three types of messages: Controller-to-Switch, Asynchronous, and Symmetric. Each message contains more subtypes.



- Controller-to-Switch messages:
 - Features message: After an SSL/TCP session is established, the controller sends Features messages to a switch to request switch information. The switch must send a response, including the interface name, MAC address, and interface rate.
 - Configuration message: The controller can set or query the switch status.
 - Modify-State message: The controller sends this message to a switch to manage the switch status, that is, to add, delete, or modify the flow table and set interface attributes of the switch.
 - Read-State message: The controller sends this message to collect statistics on the switch.
 - Send-Packet message: The controller sends the message to a specific interface of the switch.
- Asynchronous messages:
 - Packet-in message: If no matching entry exists in the flow table or the action "send-to-controller" is matched, the switch sends a packet-in message to the controller.
 - Packet-out message: The controller sends this message to respond to a switch.
 - Flow-Removed message: When an entry is added to a switch, the timeout interval is set. When the timeout interval is reached, the entry is deleted. The switch then sends a Flow-Removed message to the controller. When an entry in the flow table needs to be deleted, the switch also sends this message to the controller.
 - Port-status message: A switch sends this message to notify the controller when the interface configuration or state changes.

- Symmetric messages:
 - Hello message: When an OpenFlow connection is established, the controller and switch immediately send an OFPT_HELLO message to each other. The version field in the message is filled with the latest OpenFlow version supported by the sender. After receiving the message, the receiver calculates the protocol version number, that is, selects the smaller one between the versions supported by the sender and the receiver. If the receiver supports the version, connection requests are processed until the connection is successful. Otherwise, the receiver replies with an OFPT_ERROR message, in which the type field is filled with `ofp_error_type.OFPET_HELLO_FAILED`.
 - Echo message: Either a switch or controller can send an Echo Request message, but the receiver must reply with an Echo Reply message. This message can be used to measure the latency and connectivity between the controller and switch. That is, Echo messages are heartbeat messages.
 - Error message: When a switch needs to notify the controller of a fault or error, the switch sends an Error message to the controller.
- The OpenFlow protocol is still being updated. For more message types, see the OpenFlow Switch Specification released by Open Networking Foundation (ONF).



Flow Table Overview

- OpenFlow switches forward packets based on flow tables.
- Each flow entry includes the Match Fields, Priority, Counters, Instructions, Timeouts, Cookie, and Flags.
The Match Fields and Instructions are key fields for packet forwarding.
 - The Match Fields is a field against which a packet is matched and can be customized.
 - The Instructions field indicates OpenFlow processing when a packet matches a flow entry.

Match Fields	Priority	Counters	Instructions	Timeouts	Cookie	Flags
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Flow table fields can be customized. The following table is an example.

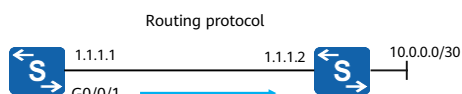
Ingress Port	Ether Source	Ether Dst	Ether Type	VLAN ID	VLAN Priority	IP Src	IP Dst	TCP Src Port	TCP Dst Port
3	MAC1	MAC2	0x8100	10	7	IP1	IP2	5321	8080

- Match Fields: a field against which a packet is matched. (OpenFlow 1.5.1 supports 45 options). It can contain the inbound interface, inter-flow table data, Layer 2 packet header, Layer 3 packet header, and Layer 4 port number.
- Priority: matching sequence of a flow entry. The flow entry with a higher priority is matched first.
- Counters: number of packets and bytes that match a flow entry.
- Instructions: OpenFlow processing when a packet matches a flow entry. When a packet matches a flow entry, an action defined in the Instructions field of each flow entry is executed. The Instructions field affects packets, action sets, and pipeline processing.
- Timeouts: aging time of flow entries, including Idle Time and Hard Time.
 - Idle Time: If no packet matches a flow entry after Idle Time expires, the flow entry is deleted.
 - Hard Time: After Hard Time expires, a flow entry is deleted regardless of whether a packet matches the flow entry.
- Cookie: identifier of a flow entry delivered by the controller.
- Flags: This field changes the management mode of flow entries.



Comparison Between Forwarding Modes

Typical Routing Protocol: Packet Forwarding Based on Routing Tables



Routing table	Destination Network	Protocol	Next Hop	Outbound Interface
	10.0.0.0/30	OSPF	1.1.1.2	G0/0/1

- In typical cases, network devices query routing tables to guide traffic forwarding.
- Entries in a routing table are calculated by running a routing protocol between network devices.
- The length of the routing table is fixed. Network devices forward packets based on the longest match rule. A network device has only one routing table.

OpenFlow: Packet Forwarding Based on Flow Tables



Flow table	Match Fields	Priority	Counters	Instructions	Timeouts	Cookie

- OpenFlow is a network protocol. Switches running OpenFlow forward traffic based on flow tables.
- Flow tables are calculated by the OpenFlow controller and then delivered to switches.
- A flow table has variable length and defines various matching and forwarding rules. A network device has multiple flow tables.

- For tables 0-255, table 0 is first matched. In a flow table, flow entries are matched by priority. The flow entry with a higher priority is matched first.
- Currently, OpenFlow is mainly used on software switches, such as OVSs and CE1800Vs, in DCs, but not on physical switches to separate forwarding and control planes.



Essential Requirements of SDN

- The essence of SDN is to make networks more open, flexible, and simple. It builds a centralized brain for a network and implements fast service deployment, traffic optimization, or network service openness through centralized control in the global view.
- SDN has the following benefits:
 - Provides centralized management, simplifying network management and O&M.
 - Shields technical differences, simplifies network configuration, and reduces O&M costs.
 - Offers automatic optimization, improving network utilization.
 - Deploys services rapidly, shortening the service rollout time.
 - Builds an open network, supporting open and programmable third-party applications.

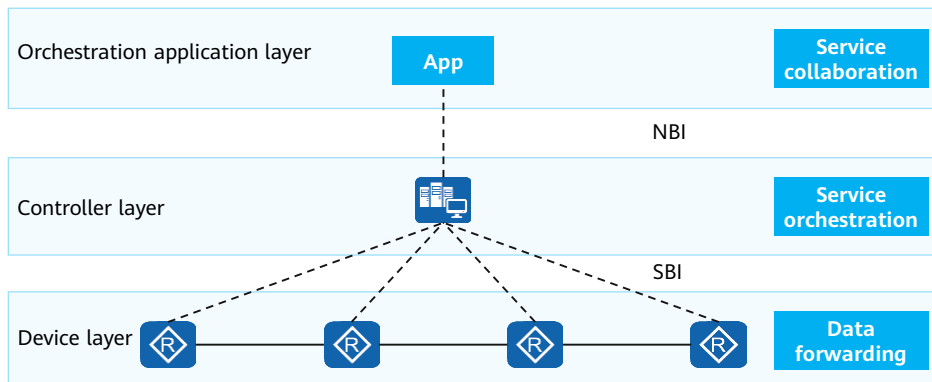
SDN transforms network architecture.

- Forwarding-control separation is a method to implement SDN.



SDN Network Architecture

- The SDN network architecture consists of the orchestration application layer, controller layer, and device layer. Different layers are connected through open interfaces. From the perspective of the controller layer, SBIs oriented to the device layer and NBIs oriented to the orchestration application layer are distinguished. OpenFlow is one of SBI protocols.

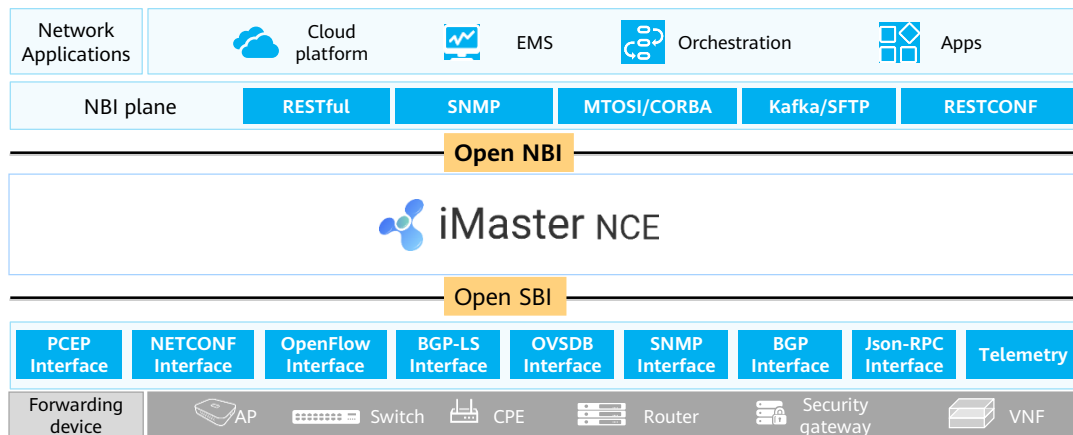


- Orchestration application layer:** provides various upper-layer applications for service intents, such as OSS and OpenStack. The OSS is responsible for service orchestration of the entire network, and OpenStack is used for service orchestration of network, compute, and storage resources in a DC. There are other orchestration-layer applications. For example, a user wants to deploy a security app. The security app is irrelevant to the user host location but invokes NBIs of the controller. Then the controller delivers instructions to each network device. The command varies according to the SBI protocol.
- Controller layer:** The SDN controller is deployed at this layer, which is the core of the SDN network architecture. The controller layer is the brain of the SDN system, and its core function is to implement network service orchestration.
- Device layer:** A network device receives instructions from the controller and performs forwarding.
- NBI:** NBIs are used by the controller to interconnect with the orchestration application layer, mainly RESTful.
- SBI:** SBIs used by the controller to interact with devices through protocols such as NETCONF, SNMP, OpenFlow, and OVSDB.



Huawei SDN Network Architecture

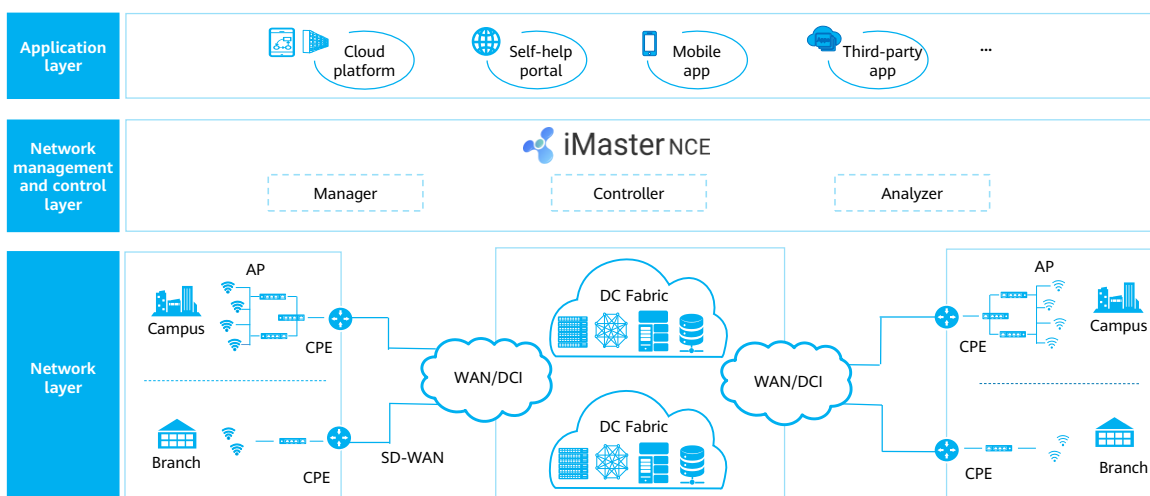
- Huawei SDN network architecture supports various SBIs and NBIs, including OpenFlow, OVSDb, NETCONF, PCEP, RESTful, SNMP, BGP, JSON-RPC, and RESTCONF interfaces.



- Cloud platform: resource management platform in a cloud DC. The cloud platform manages network, compute, and storage resources. OpenStack is the most mainstream open-source cloud platform.
- The Element Management System (EMS) manages one or more telecommunication network elements (NEs) of a specific type.
- Orchestration (container orchestration): The container orchestration tool can also provide the network service orchestration function. Kubernetes is a mainstream tool.
- MTOSI or CORBA is used to interconnect with the BSS or OSS. Kafka or SFTP can be used to connect to a big data platform.



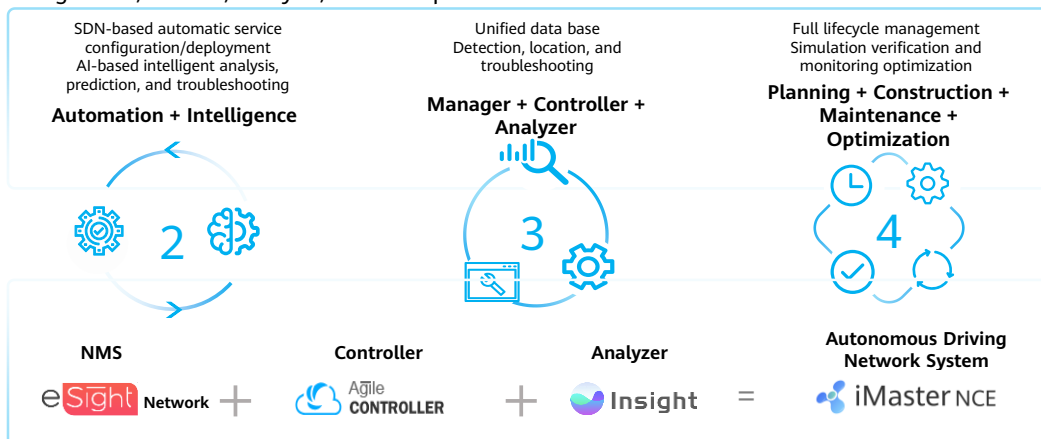
Huawei SDN Solution - Integrating Management, Control, and Analysis to Build an Intent-Driven Network





Introduction to iMaster NCE

- Huawei iMaster NCE is the industry intelligent network automation platform that integrates management, control, analysis, and AI capabilities.



- iMaster NCE converts service intents into physical network configurations. It manages, controls, and analyzes global networks in a centralized manner in the southbound direction. It enables resource cloudification, full-lifecycle network automation, and intelligent closed-loop driven by data analysis for business and service intents. It provides northbound open APIs for quick integration with IT systems.
- iMaster NCE can be used in the enterprise data center network (DCN), enterprise campus, and enterprise branch interconnection (SD-WAN) scenarios to make enterprise networks simple, smart, open, and secure, accelerating enterprise service transformation and innovation.



iMaster NCE Application



DC

iMaster NCE-Fabric *



Enterprise
campus

iMaster NCE-Campus *



SD-WAN

iMaster NCE-WAN



IP WAN

iMaster NCE-IP

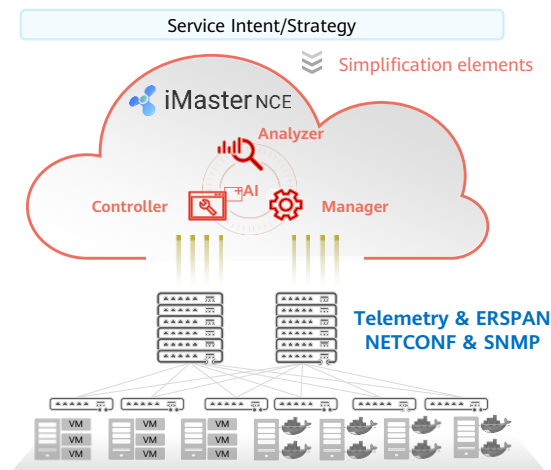


WAN
transmission

iMaster NCE-T

* iMaster NCE-Fabric and iMaster NCE-Campus are introduced in this document.

- Based on iMaster NCE-Fabric, DCNs provide full-lifecycle services from planning, construction, O&M, to optimization.



- The planning tool interconnects with iMaster NCE-Fabric to implement integrated planning and construction.
- Zero Touch Provisioning (ZTP)

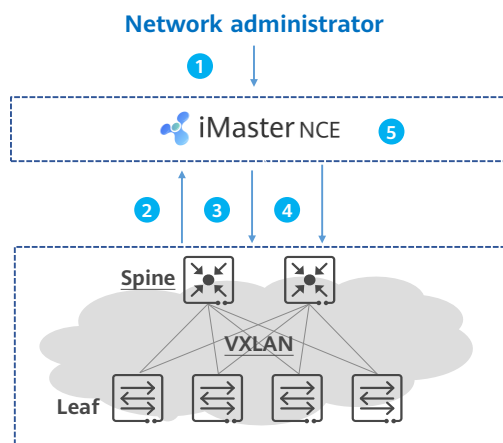
- Service intent self-understanding and conversion
- Network change simulation and evaluation, eliminating human errors

- Rapid fault detection and location based on knowledge graph and expert experience
- Fast fault rectification based on expert experiences and simulation analysis

- AI-Fabric-oriented local traffic inference and online model training and optimization
- User behavior prediction and resource optimization suggestions



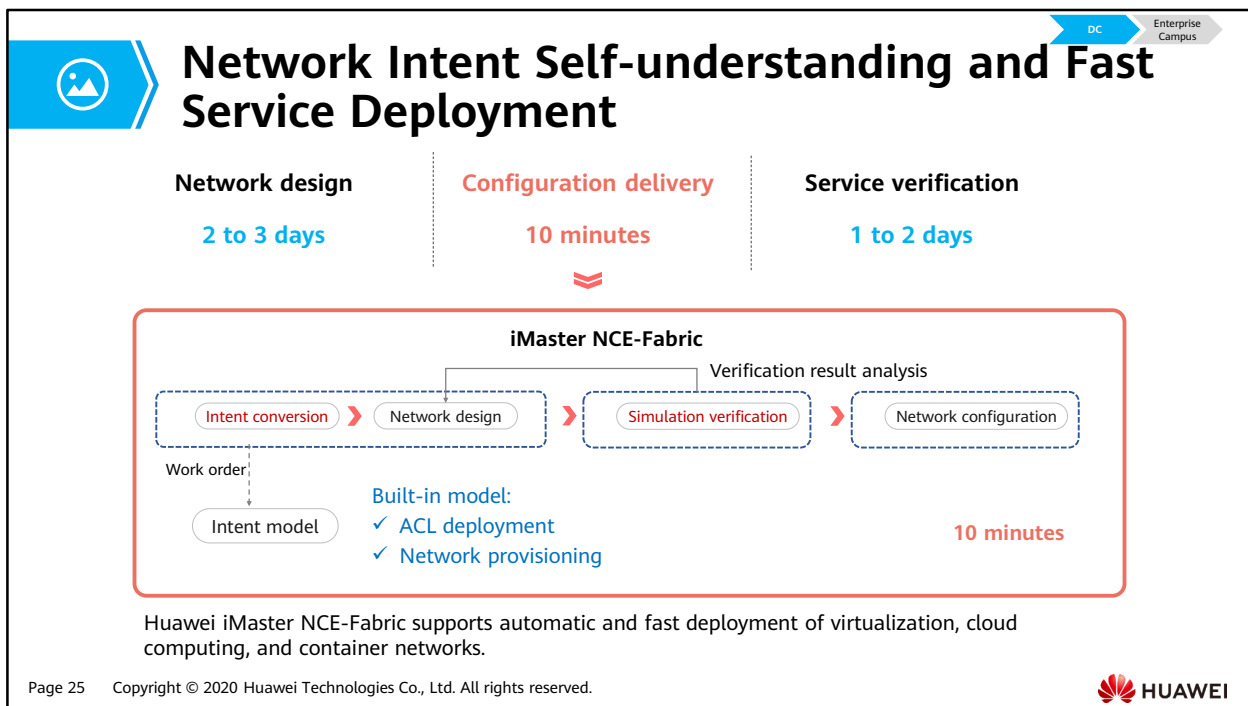
Simplified ZTP Deployment



ZTP deployment process:

1. The network administrator clicks the icon on iMaster NCE to start the ZTP task.
2. A device automatically obtains an IP address to access iMaster NCE.
3. iMaster NCE determines the device role (spine or leaf node), delivers configurations such as the management IP address, SNMP configuration, and NETCONF configuration to online devices, and manages the devices through the management IP address.
4. iMaster NCE globally delivers interconnection configurations as well as OSPF or BGP configurations.
5. The device goes online successfully, and the administrator views network-wide information on iMaster NCE.

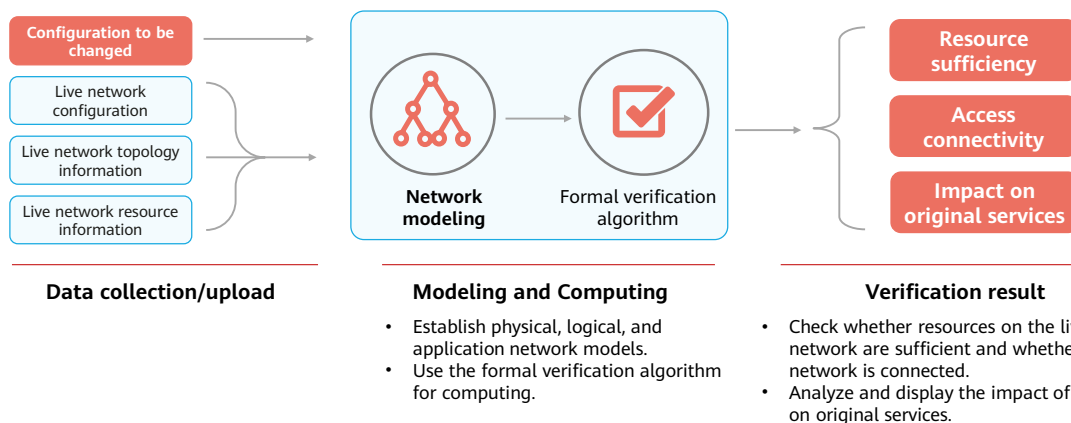
Note: The DC uses the spine-leaf architecture.

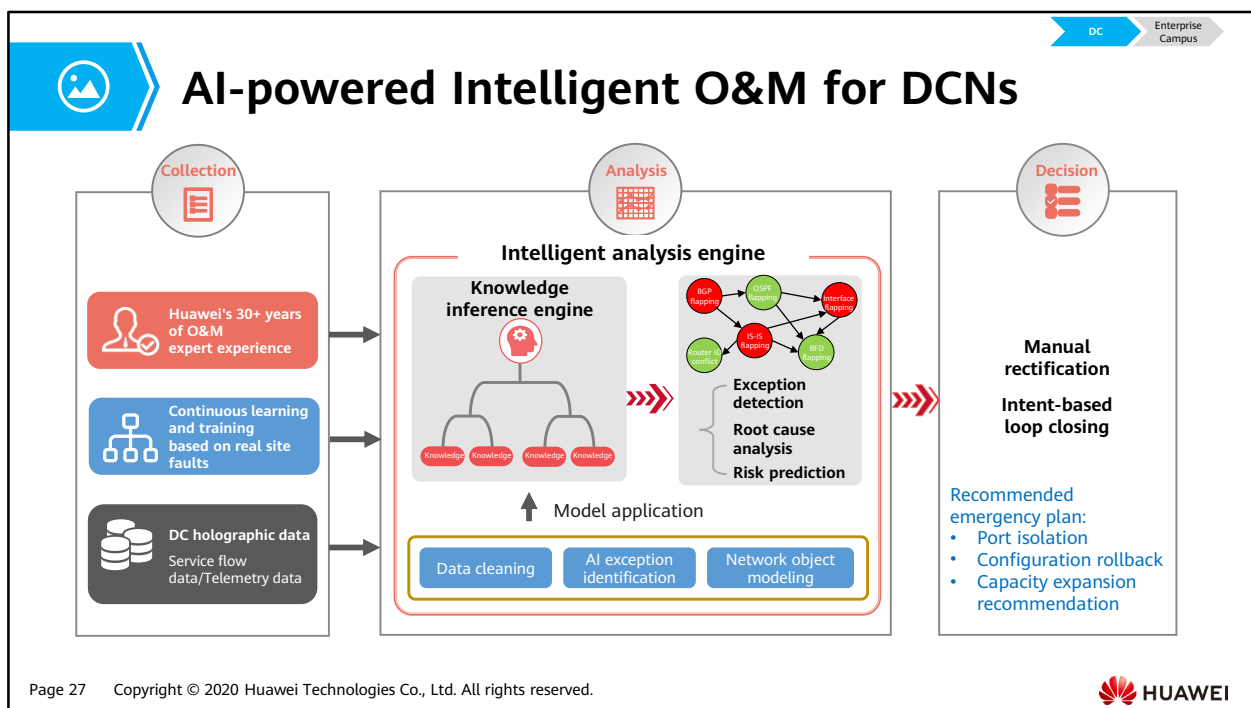


- iMaster NCE-Fabric can connect to a user's IT system to match the intent model for user intents and deliver configurations to devices through NETCONF to implement fast service deployment.
- iMaster NCE-Fabric can interconnect with the mainstream cloud platform (OpenStack), virtualization platform (vCenter/System Center), and container orchestration platforms (Kubernetes).



Network Change Simulation and Change Risk Prediction

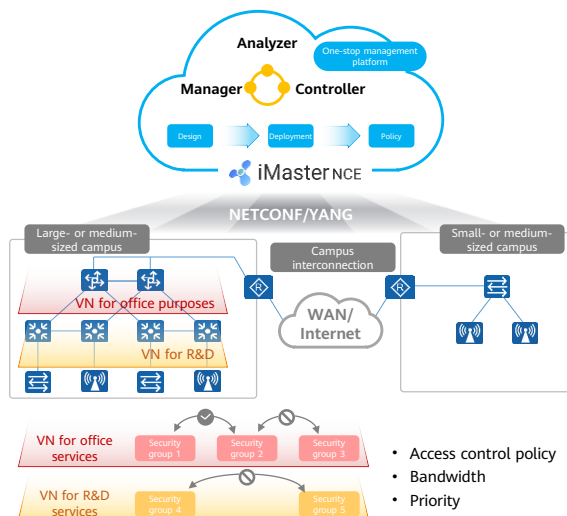




- iMaster NCE-FabricInsight provides AI-based intelligent O&M capabilities for DCs.



Huawei CloudCampus Autonomous Driving Network Solution



Fast network deployment, improving deployment efficiency by 600%

- Device plug-and-play: simplified device deployment, scenario navigation, and template-based configuration
- Simplified network deployment: Network resource pooling, multi-purpose network, and automatic service provisioning

Fast service provisioning, improving user experience by 100%

- Free mobility: GUI-based policy configuration, allowing users to access the network anytime and anywhere without changing the roaming permission and user experience
- Intelligent terminal identification: Anti-spoofing for terminal access, with an intelligent terminal identification accuracy of over 95%
- Intelligent HQoS: Application-based scheduling and shaping, and refined bandwidth management, ensuring service experience of key users

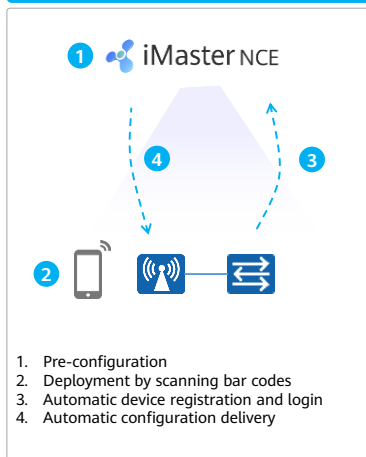
Fast intelligent O&M, improving network performance by over 50%

- Real-time experience visualization: Telemetry-based network experience visualization at each moment, for each user, and in each area
- Precise fault analysis: Proactively identifying 85% of typical network issues and providing suggestions, and comparing and analyzing real-time data to predict faults
- Intelligent network optimization: Predictive optimization of wireless networks based on historical data, improving network-wide performance by over 50% (Source: Tolly Certification)

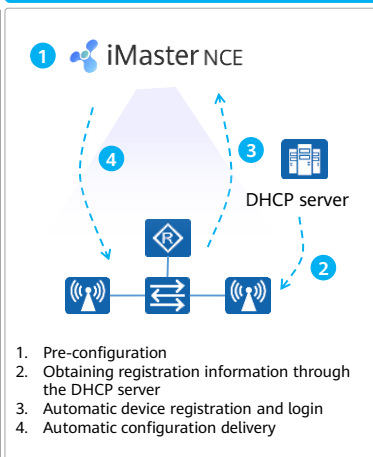


Device Plug-and-Play

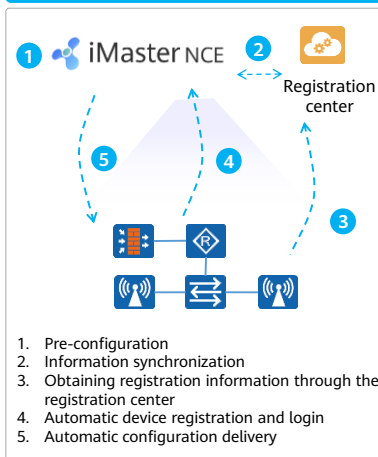
Deployment by Scanning Bar Codes



DHCP-based Deployment



Deployment Through the Registration Center

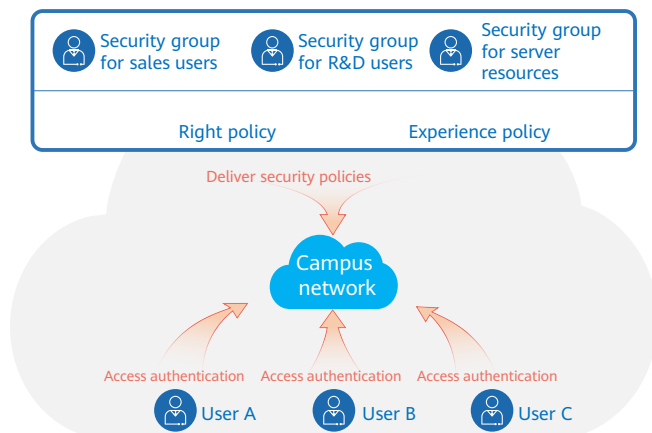


- Device plug-and-play includes but is not limited to deployment by scanning bar codes using an app, DHCP-based deployment, and deployment through the registration query center.
- Registration center: Huawei device registration query center, also called registration center, is one of the main components of Huawei CloudCampus solution. It is used to query the device management mode and registration ownership. A device determines whether to switch to the cloud-based management mode and which cloud management platform to register with based on the query result. The AP is used as an example. Huawei devices that support cloud-based management are pre-configured with the URL (register.naas.huawei.com) and port number (10020) of the Huawei device registration center.



Free Mobility: Policy Management Based on Security Groups

- Free mobility: Enables users to have consistent network rights and security policies regardless of their locations and IP addresses.

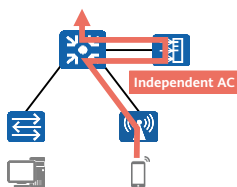


- 1 Use security groups. A security group is a group of users for which the same security policy is used.
- 2 Define security group-based permission control policies and user experience policies and deliver the policies to network devices.
- 3 A security group is authorized to a user after the user passes access authentication.
- 4 After user traffic enters a network, network devices enforce policies based on the source and destination security groups of the traffic.



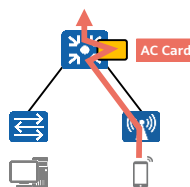
Wired and Wireless Convergence

WLAN Construction Mode 1: Standalone AC



- This mode poses a wireless traffic bottleneck and increases faulty nodes.
- Wired and wireless management is independent.
- Wired and wireless authentication points are separated.

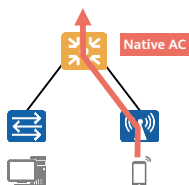
WLAN Construction Mode 2: AC Card



- An AC card is installed on a switch to provide AC functions.
- Hardware-level convergence.

Wired and wireless authentication point separation, distributed policy control, separation of control and data traffic forwarding, and troubleshooting and management difficulties.

Wired and Wireless Convergence (Native AC)



The switch integrates the AC function, eliminating wireless traffic forwarding bottlenecks and reducing fault nodes. Wired and wireless devices are centrally managed.

- Unified management and converged forwarding of wired and wireless services
- Converged management for wired and wireless users and gateway convergence
- Converged authentication points for wired and wireless access
- Unified wired and wireless policy execution



Intelligent Terminal Identification, Ensuring Secure Access

Requirements and Challenges

A university

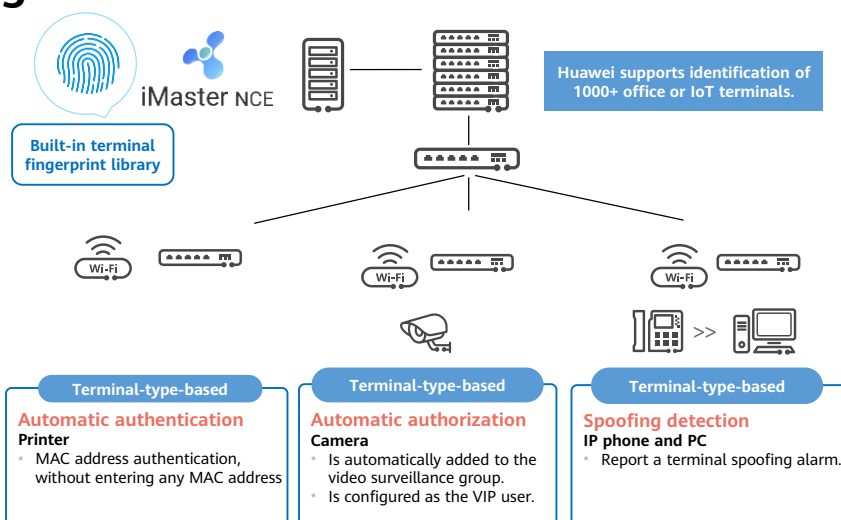
50+ types of smart terminals

Terminal information is collected by IT departments of colleges: MAC address collection is difficult and error-prone.

An enterprise

100+ authentication faults reported per day

It is difficult to locate access spoofing.





HQoS: User- and Application-based QoS Policy

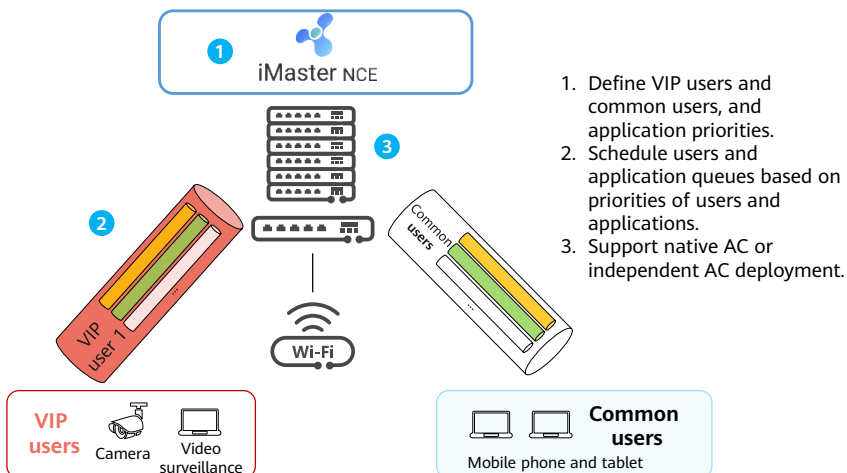
User- and application-based QoS policies ensure experience of high-priority users and applications

Requirements and Challenges

Traditional QoS policies are invalid for video services.

(Example) Building surveillance scenario: Wireless video services of common users increase, occupying a large number of network resources and causing network congestion.

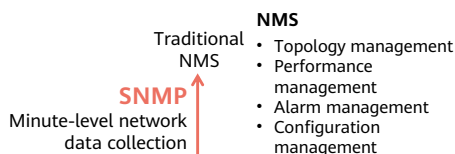
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AI-Powered Intelligent O&M of Campus Networks

As-Is: Device-Centered Network Management



- **Device-centric O&M method:** User experience cannot be detected.
- **Fault-triggered responses:** Potential faults cannot be identified.
- Rely on professional engineers to **locate faults onsite**.



To-Be: User Experience-Centered AI-Powered Intelligent O&M



Telemetry

Second-level network data collection

Intelligent network analyzer

- Visualized user experience management
- User journey playback
- Potential fault identification
- Root cause identification
- Predictive network optimization



Experience visualization: Telemetry-based second-level data collection, visualized experience of each user and each application in real time

Minute-level potential fault identification and root cause location

- Identify potential faults based on dynamic baselines and big data correlation analysis.
- Accurately locate root causes of faults through KPI association analysis and protocol tracing.

Predictive network optimization: AI is used to intelligently analyze the AP load trend and implement predictive optimization on the wireless network.

The efficiency is improved by using algorithms. With scenario-based continuous learning and expert experience, intelligent O&M frees O&M personnel from complex alarms and noises, making O&M more automated and intelligent.



AI-Powered Intelligent Radio Calibration

Traditional radio calibration cannot achieve expected results

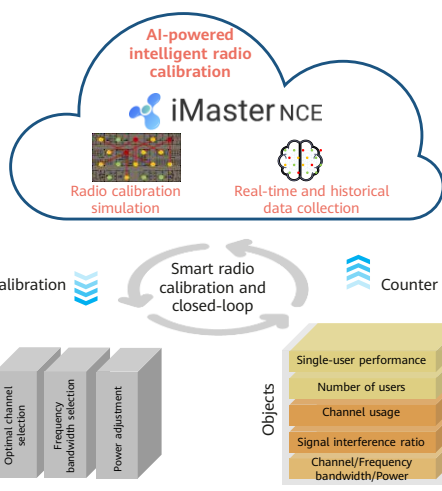
Phase 1: Manual Calibration

Based on engineers' experience, the calibration is time-consuming and error-prone, and the calibration result is unstable.



Phase 2: Automatic Calibration

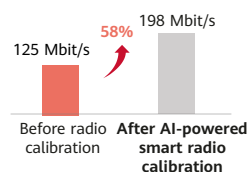
Calibration based on real-time radio interference does not consider device load changes, and the calibration result cannot be ensured.



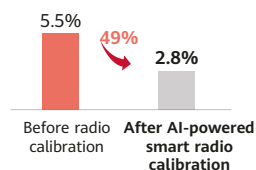
Tested and verified by authoritative organizations



Average downlink rate per terminal:



Average Wi-Fi channel interference





Contents

1. SDN Overview
- 2. NFV Overview**

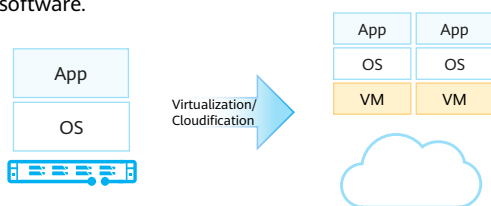


NFV Background: Thinking from IT Industry Transformation

- The IT industry transformation brings thinking on **network architecture** and **device architecture** in the network industry. The network architecture layer involves the SDN controller and the device architecture layer involves the device deployment mode.

IT Industry Transformation

- In recent years, IT technologies such as virtualization and cloud computing have been booming, and applications deployed on hardware have been gradually migrated to the cloud. Applications are deployed on private clouds, public clouds, or hybrid clouds as software.



Network?

- Thinking about the network industry: Can network applications be deployed in a software-based manner?
- In the context, Network Functions Virtualization (NFV) is introduced.

- Virtualized network functions (VNFs) are implemented by virtualizing traditional NEs such as IMSs and CPEs of carriers. After hardware is universalized, traditional NEs are no longer the products with embedded software and hardware. Instead, they are installed on universal hardware (NFVI) as software.



Origin of NFV

- In October 2012, 13 top carriers (including AT&T, Verizon, VDF, DT, T-Mobile, BT, and Telefonica) released the first version of NFV White Paper at the SDN and OpenFlow World Congress. In addition, the Industry Specification Group (ISG) was founded to promote the definition of network virtualization requirements and the formulation of the system architecture.
- In 2013, the ETSI NFV ISG conducted the first phase of research and completed the formulation of related standards. The ETSI NFV ISG defined NFV requirements and architecture and sorts out the standardization processes of different interfaces.



- In 2015, NFV research entered the second phase. The main research objective is to build an interoperable NFV ecosystem, promote wider industry participation, and ensure that the requirements defined in phase 1 are met. In addition, the ETSI NFV ISG specified the collaboration relationships between NFV and SDN standards and open source projects. Five working groups are involved in NFV phase 2: IFA (architecture and interface), EVE (ecosystem), REL (reliability), SEC (security), and TST (test, execution, and open source). Each working group mainly discusses the deliverable document framework and delivery plan.
- The ETSI NFV standard organization cooperates with the Linux Foundation to start the open source project OPNFV (NFV open source project, providing an integrated and open reference platform), integrate resources in the industry, and actively build the NFV industry ecosystem. In 2015, OPNFV released the first version, further promoting NFV commercial deployment.
- NFV-related standard organizations include:
 - ETSI NFV ISG: formulates NFV requirements and functional frameworks.
 - 3GPP SA5 working group: focuses on technical standards and specifications of 3GPP NE virtualization management (MANO-related).
 - OPNFV: provides an open-source platform project that accelerates NFV marketization.



NFV Value

- NFV aims to address issues such as complex deployment and O&M and service innovation difficulties due to large numbers of telecom network hardware devices. NFV brings the following benefits to carriers while reconstructing telecom networks:
 - Shortened service rollout time
 - Reduced network construction cost
 - Improved network O&M efficiency
 - Open ecosystem

- Shortened service rollout time: In the NFV architecture, adding new service nodes becomes simple. No complex site survey or hardware installation is required. For service deployment, you only need to request virtual resources (compute, storage, and network resources) and software loading, simplifying network deployment. To update service logic, you simply need to add new software or load new service modules to complete service orchestration. Service innovations become simple.
- Reduced network construction cost: Virtualized NEs can be integrated into COTS devices to reduce the cost. Enhancing network resource utilization and lowering power consumption can lower overall network costs. NFV uses cloud computing technologies and universal hardware to build a unified resource pool. Resources are dynamically allocated on demand based on service requirements, implementing resource sharing and improving resource utilization. For example, automatic scale-in and scale-out can be used to solve the resource usage problem in the tidal effect.
- Enhanced network O&M efficiency: Automated and centralized management improves the operation efficiency and reduces the O&M cost. Automation includes DC-based hardware unit management automation, MANO application service life management automation, NFV- or SDN-based coordinated network automation.
- Open ecosystem: The legacy telecom network exclusive software/hardware model defines a closed system. NFV-based telecom networks use an architecture based on standard hardware platforms and virtual software. The architecture easily provides open platforms and open interfaces for third-party developers, and allows carriers to build open ecosystems together with third-party partners.



Key NFV Technologies: Virtualization

- Virtualization is the foundation of NFV, and cloudification is the key.
- On traditional telecom networks, each NE is implemented by dedicated hardware, resulting in high costs and difficult O&M. Virtualization features partition, isolation, encapsulation, and independence from hardware, which can meet NFV requirements. Carriers use virtualization to run software-based NEs on universal infrastructures.

Partition



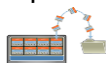
Multiple VMs can concurrently run on a single physical server.

Isolation



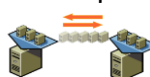
VMs that run on the same server are isolated from each other.

Encapsulation



All data of a VM is saved in files. A VM can be moved and replicated by moving and replicating the files.

Hardware independence



VMs can run on any servers without any modifications.

- On traditional telecom networks, each NE is implemented by dedicated hardware. A large number of hardware interoperability tests, installation, and configuration are required during network construction, which is time-consuming and labor-consuming. In addition, service innovation depends on the implementation of hardware vendors, which is time-consuming and cannot meet carriers' service innovation requirements. In this context, carriers want to introduce the virtualization mode to provide software NEs and run them on universal infrastructures (including universal servers, storage devices, and switches).
- Using universal hardware helps carriers reduce the cost of purchasing dedicated hardware. Service software can be rapidly developed through iteration, which enables carriers to innovate services quickly and improve their competitiveness. By doing this, carriers can enter the cloud computing market.



Key NFV Technology: Cloudification

- As defined by the National Institute of Standards and Technology (NIST), cloud computing is a model that allows users to obtain resources (for example, networks, servers, storage devices, applications, services) in a shared compute resource pool based on their needs anytime, anywhere. This model enables fast resource provisioning and release, and minimizes the resource management workload and interactions with service providers.
- Cloud computing has many advantages. Cloudification of network functions on carriers' networks mainly uses resource pooling and rapid elastic scaling.

Characteristics of Cloud Computing



- According to the NIST, cloud computing services have the following characteristics:
 - On-demand self-service: Cloud computing implements on-demand self-service of IT resources. Resources can be requested and released without intervention of IT administrators.
 - Broad network access: Users can access networks anytime and anywhere.
 - Resource pooling: Resources including networks, servers, and storage devices in a resource pool can be provided for users.
 - Rapid elasticity: Resources can be quickly provisioned and released. The resource can be used immediately after being requested, and can be reclaimed immediately after being released.
 - Measured service: The charging basis is that used resources are measurable. For example, charging is based on the number of CPUs, storage space, and network bandwidth.



Introduction to the NFV Architecture

- The NFV architecture includes the network functions virtualization infrastructure (NFVI), a virtualized network function (VNF), and management and orchestration (MANO). In addition, the NFV architecture needs to support the existing business support system (BSS) or operations support system (OSS).

OSS/BSS: is an existing operation/O&M support system.

VNF: uses cloud resources to construct software NEs.

NFVI: provides cloud-based resource pools.

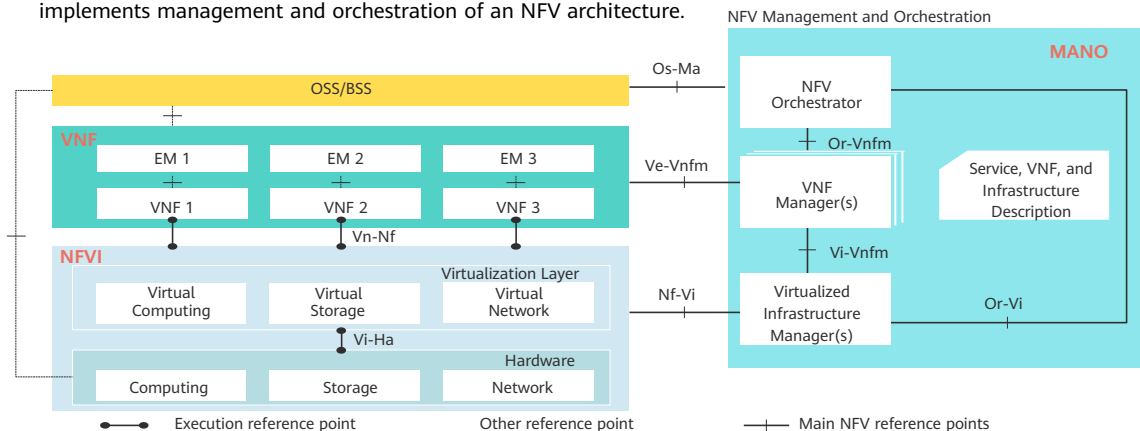
MANO:
Provides functions such as service orchestration, service management, and resource management.

- Each layer of the NFV architecture can be provided by different vendors, which improves system development but increases system integration complexity.
- NFV implements efficient resource utilization through device normalization and software and hardware decoupling, reducing carriers' TCO, shortening service rollout time, and building an open industry ecosystem.
- The NFVI consists of the hardware layer and virtualization layer, which are also called COTS and CloudOS in the industry.
 - COTS: universal hardware, focusing on availability and universality, for example, Huawei FusionServer series hardware server.
 - CloudOS: cloud-based platform software, which can be regarded as the operating system of the telecom industry. CloudOS virtualizes physical compute, storage, and network resources into virtual resources for upper-layer software to use, for example, Huawei FusionSphere.
- VNF: A VNF can be considered as an app with different network functions and is implemented by software of traditional NEs (such as IMS, EPC, BRAS, and CPE) of carriers.
- MANO: MANO is introduced to provision network services in the NFV multi-CT or multi-IT vendor environment, including allocating physical and virtual resources, vertically streamlining management layers, and quickly adapting to and interconnecting with new vendors' NEs. The MANO includes the Network Functions Virtualization Orchestrator (NFVO, responsible for lifecycle management of network services), Virtualized Network Function Manager (VNFM, responsible for lifecycle management of VNFs), and Virtualized Infrastructure Manager (VIM, responsible for resource management of the NFVI).



Standard NFV Architecture

- ETSI defines the standard NFV architecture, which consists of the NFVI, VNF, and MANO. The NFVI includes the universal hardware layer and virtualization layer. The VNF is implemented using software, and the MANO implements management and orchestration of an NFV architecture.





Functional Modules of the NFV Architecture

- Main functional modules defined in the standard NFV architecture:

OSS or BSS

Management system for a service provider. It is not a functional component in the NFV architecture, but the MANO must provide an interface for interoperation with the OSS or BSS.

MANO

NFV management and orchestration. The MANO includes the VIM, VNFM, and NFVO, and provides unified management and orchestration for VNFs and the NFVI.

- **VIM:** NFVI management module that runs on an infrastructure site. The VIM provides functions such as resource discovery, virtual resource management and allocation, and fault handling.
- **VNFM:** It controls the VNF lifecycle (including instantiation, configuration, and shutdown).
- **NFVO:** It orchestrates and manages all the software resources and network services on an NFV network.

VNF

VNFs refer to VMs as well as service NEs and network function software deployed on the VMs.

NFVI

NFV infrastructure, including required hardware and software. The NFVI provides a running environment for VNFs.

- **Hardware layer:** includes hardware devices that provide compute, network, and storage resources.
- **Virtualization layer:** abstracts hardware resources to form virtual resources, such as virtual compute, storage, and network resources. The virtualization function is implemented by Hypervisor^[1].

- BSS: business support system
- OSS: operation support system
- A hypervisor is a software layer between physical servers and OSs. It allows multiple OSs and applications to share the same set of physical hardware. It can be regarded as a meta operating system in the virtual environment, and can coordinate all physical resources and VMs on the server. It is also called virtual machine monitor (VMM). The hypervisor is the core of all virtualization technologies. Mainstream hypervisors include KVM, VMWare ESXi, Xen, and Hyper-V.



NFV Architecture Interfaces

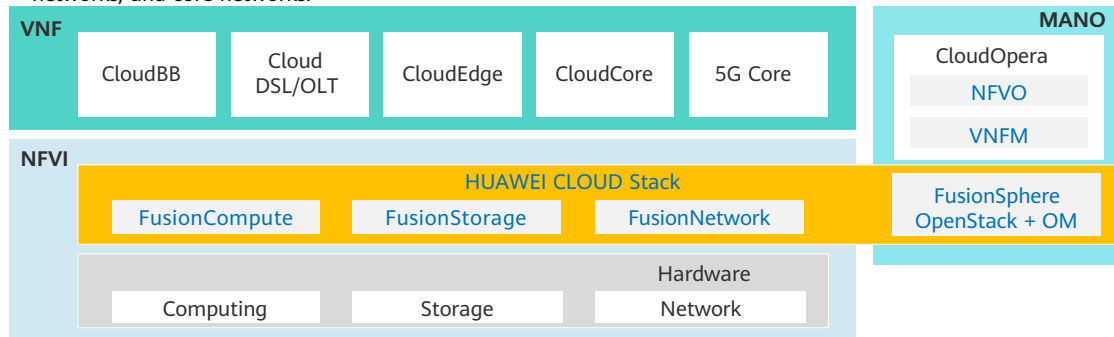
- Main interfaces of the standard NFV architecture:

Interface	Description
Vi-Ha	Is used between the virtualization layer and hardware layer. The virtualization layer meets basic hardware compatibility requirements.
Vn-Nf	Is used between a VM and the NFVI. It ensures that VMs can be deployed on the NFVI to meet performance, reliability, and scalability requirements. The NFVI meets VMs' OS compatibility requirements.
Nf-Vi	Is used between the virtualization layer management software and NFVI. It provides management of virtual computing, storage, and network systems of NFVI, virtual infrastructure configuration and connections, as well as system usage, performance monitoring, and fault management.
Ve-Vnfm	Is used between the VNFM and a VNF, implementing VNF lifecycle management, VNF configuration, VNF performance, and fault management.
OS-Ma	Manages lifecycles of network services and VNFs.
Vi-Vnfm	Is used for interaction between the service application management system or service orchestration system and virtualization layer management software.
Or-Vnfm	Sends configuration information to the VNFM, configures the VNFM, and connects the orchestrator and VNFM. It exchanges information with the NFVI resources allocated to VNFs and information between VNFs.
Or-Vi	Is used to send resource reservation and resource allocation requests required by the orchestrator and exchange virtual hardware resource configurations and status information.



Huawei's NFV Solution

- In the Huawei NFV architecture, functions of the virtualization layer and VIM are implemented by the HUAWEI CLOUD Stack NFVI platform. HUAWEI CLOUD Stack can virtualize compute, storage, and network resources and centrally manage, monitor, and optimize physical virtualization resources.
- Huawei provides cloud-based solutions for carriers' wireless networks, bearer networks, transport networks, access networks, and core networks.



- DSL: Digital Subscriber Line
- OLT: Optical Line Terminal



FAQ

- Q1: What is the relationship between SDN and NFV in the industry?
- A: Both SDN and NFV involve network transformation and the NFV concept was proposed at the SDN and OpenFlow World Congress. However, they are independent of each other. SDN mainly affects the network architecture, and NFV mainly affects the NE deployment mode.
- Q2: What is the relationship between SDN and NFV in Huawei solutions?
- A: Huawei provides different solutions for SDN and NFV, but they are associated. Huawei NFVI solution is provided by HUAWEI CLOUD Stack.



Quiz

1. (Multiple) Which of the following statements about Huawei SDN solution are true? ()
 - A. The solution supports various SBI protocols, such as RESTful, NETCONF, and OVSDB.
 - B. OpenFlow can be used as the SBI protocol.
 - C. The solution integrates management, control, and analysis to build a simplified network.
 - D. The solution provides open and programmable network interfaces to support third-party application development and system interconnection.
2. Please briefly describe the benefits of NFV.

1. BCD
2. NFV aims to address issues such as complex deployment and O&M and service innovation difficulties due to large numbers of telecom network hardware devices. NFV brings the following benefits to carriers while reconstructing telecom networks:
 - Shortened service rollout time
 - Reduced network construction cost
 - Improved network O&M efficiency
 - Open ecosystem



Summary

- With the transformation and development of the network industry, SDN and NFV are proposed.
- SDN is an innovation of network architecture. It uses a controller to make networks more open, flexible, and simple.
- NFV is an innovation in the deployment of telecom network devices. Based on virtualization and cloud computing, NFV helps reconstruct telecom networks.



More Information

- For more information about OpenFlow, visit <https://www.opennetworking.org/>.
- For more information about Huawei SDN solution, see the HCIP course.



Thank You
www.huawei.com